Plainpaper

Academic papers, in plain English.



Hackathon 3
Equal Access

SDG 10
Reduced Inequalities

Tool
Gemini API

Built by
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CONTEXT

The Problem

Who is affected?

IT bachelor students in countries where curricula are outdated, still teaching legacy frameworks and not covering modern research papers.

What is the barrier?

Academic papers use dense jargon, passive voice, and assume background knowledge these students were never taught.

Why does it matter?

The papers are free (arXiv, open access), but the language shuts out exactly the students who need them most.

Bass & Heeks (2011): computing curricula in many African universities date from the mainframe era

CONTEXT

The Numbers

91%

more time needed by non-native speakers to read a paper

Amano et al., 2023, PLOS Biology

2.5x

more likely to have papers rejected for language quality alone

Amano et al., 2023, PLOS Biology

1.5 to 2.5x

more correct arguments when students learn concepts in plain language first

McDonnell et al., 2016

78%

of students agreed it is easier to learn concepts in everyday language before jargon

McDonnell et al., 2016

CONTEXT

Who Is This For?

PRIMARY **Primary user**	IT/CS bachelor students at universities with outdated curricula and limited access to modern materials
SECONDARY **Secondary user**	Non-native English speaking IT students in EU universities
OUT OF SCOPE **Not for**	K-12 students, non-IT majors, anyone needing native-language translation, medical/legal professionals

THE SOLUTION

Plainpaper

What it does

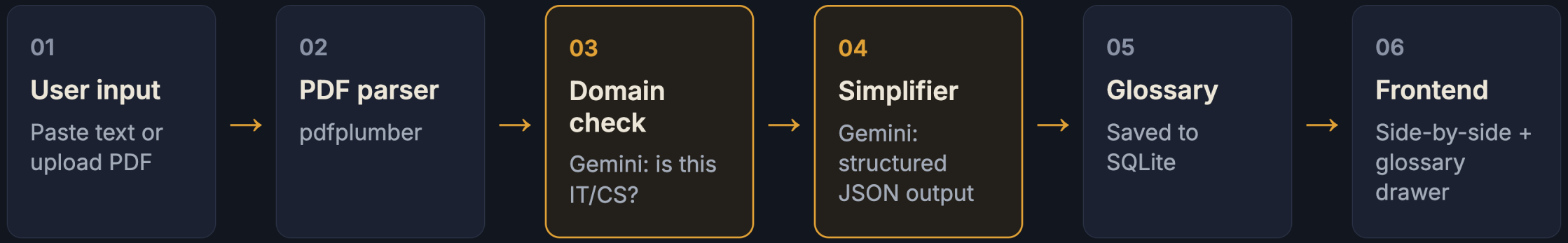
- Rewrites prose in plain English, section by section
- Preserves math and algorithms untouched
- Side-by-side view: original vs simplified
- Persistent glossary bank
- Domain guard warns on non-IT papers

What it does NOT do

- Summarize papers
- Explain math or algorithms
- Replace reading the original

THE SOLUTION

How It Works



Gemini API call

FastAPI + Vue.js + SQLite + Gemini API

THE SOLUTION

Why Not Just Use ChatGPT?

01

No prompting skill required

System prompt encodes the right behavior automatically

02

Side-by-side verification

Built into the UX, not a manual step

03

Persistent glossary bank

Accumulates across papers and sessions

04

Consistent structured output

Same format every time, regardless of user input

ETHICS

Risks

RISK 01

Silent meaning distortion

AI may change meaning without the student noticing.

e.g.
"may correlate" → "causes"

RISK 02

Over-reliance

Students may skip reading the original entirely.

RISK 03

Language bias

Simplifies to English only, so students with weak English get false confidence.

ETHICS

How We Address Them

FOR RISK 01

Side-by-side view for spot-checking; math left untouched

FOR RISK 02

Original text shown alongside, not hidden; full structure preserved

FOR RISK 03 Unmitigated

Scoped to working English proficiency. Future work: multi-language support.

REMAINING GAP No automated detection of wrong simplifications yet

WRAP-UP

Summary

PROBLEM	Jargon-heavy papers shut out students at under-resourced universities
SOLUTION	Plain English rewriting with verification and glossary
TECH	FastAPI + Vue.js + Gemini API + SQLite
DIFFERENTIATOR	Not a ChatGPT wrapper: real product features
ETHICS	3 risks identified, 2 mitigated, 1 acknowledged